Mid Examination-I   Tutorial Sheet (ECE/CSE)

Long Answer Questions

1) a) Determine the rank of the Matrix by reducing to the Echelon form $\begin{bmatrix} 1 & 2 & 3 & 0 \\ 2 & 4 & 3 & 2 \\ 3 & 2 & 1 & 3 \\ 6 & 8 & 7 & 5 \end{bmatrix}$

   b) Determine the rank of the Matrix $\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$ by reducing to the Normal form

2) Test for consistency and solve the following system of equations x + 2y + 2z = 4, 2x - y + 3z = 9, 3x - y -z = 2

3) Investigate the values of λ and μ so that the equations 2x + 3y + 5z = 9,7x + 3y – 2z = 8, 2x + 3y + λz = μ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

4) Solve the Equations 10x+y-z=11.19, x+10y+z=28.08, -x+y+10z=35.61 by Gauss-Jacobi Iterative Method.

5) Solve the equations 4x + 2y + z + 3w = 0, 6x + 3y + 4z + 7w = 0, 2x + y + w = 0.

6) Verify Cayley Hamilton theorem for the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ and also find A^{-1} & A^4

7) Determine the Eigen values and Eigen vectors for the Matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

8) Reduce the quadratic form $3x^2 +3y^2 +3z^2 +2xy +2xz -2yz$ to canonical form. Also determine the Rank, index, signature& nature

9) (a) State Rolle’s theorem and verify Rolle’s Theorem for $f(x) = \frac{sinx}{e^x}$ in $(0, \pi)$.

   (b) State Rolle’s theorem and verify Rolle’s theorem for $f(x) = (x-a)^m(x-b)^n$ Where m, n are positive integers [a, b].

10) (a)State Lagrange’s Mean value theorem and Verify Lagrange’s mean value theorem for $f(x) = (x-1)(x-2)(x-3)$ in $(0, 4)$.

   (b) State Lagrange’s Mean value theorem and If f(x) = $\sin^{-1}(x)$, 0<a <b<1 use Mean value theorem to P.T $\frac{b-a}{\sqrt{1-a^2}} < \sin^{-1} b - \sin^{-1} a < \frac{b-a}{\sqrt{1-b^2}}$

Short Answer Questions

1) Define Echelon form of a Matrix

2) Find the Rank of the matrix A= $\begin{bmatrix} 8 & 1 & 3 \\ 6 & 0 & 3 \\ 2 & 2 & -8 \end{bmatrix}$

3) Solve $x + y + z = 6$; $3x + 3y + 4z = 20$; $2x + y + 3z = 13$ by Gauss elimination method

4) When does a homogeneous system AX=O possess trivial and Non Trivial solutions.

5) Find the characteristic equation of the matrix A= $\begin{bmatrix} 1 & 3 & 5 \\ 0 & 4 & 3 \\ 0 & 0 & 7 \end{bmatrix}$

6) Find the Eigen values of A^{-1} and A^2 , if A= $\begin{bmatrix} 2 & 5 & 9 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}$

7) If two Eigen values of a matrix $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ are 3, 15 then find the third Eigen value

8) Write the Quadratic form corresponding to the symmetric matrix $\begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 3 \\ -1 & 3 & 4 \end{bmatrix}$

9) State whether Lagrange's mean value theorem can be applied to the function

$$f(x) = x^{1/3} \text{ in } (-1,1). \text{justify your answer}$$

10) State Rolle's Theorem.